

```
!pip install chromadb sentence-transformers
```

```
Requirement already satisfied: orjson>=3.9.12 in /usr/local/lib/py
Requirement already satisfied: httpx>=0.27.0 in /usr/local/lib/pytl
Requirement already satisfied: rich>=10.11.0 in /usr/local/lib/pytl
Requirement already satisfied: jsonschema>=4.19.0 in /usr/local/lib/pytl
Requirement already satisfied: transformers<6.0.0,>=4.41.0 in /usr/
Requirement already satisfied: huggingface-hub>=0.20.0 in /usr/loca
Requirement already satisfied: torch>=1.11.0 in /usr/local/lib/pytl
Requirement already satisfied: scikit-learn in /usr/local/lib/pytho
Requirement already satisfied: scipy in /usr/local/lib/python3.12/c
Requirement already satisfied: packaging>=24.0 in /usr/local/lib/py
Requirement already satisfied: pyproject_hooks in /usr/local/lib/py
Requirement already satisfied: anyio in /usr/local/lib/python3.12/c
Requirement already satisfied: certifi in /usr/local/lib/python3.12/c
Requirement already satisfied: httpcore==1.* in /usr/local/lib/pytl
Requirement already satisfied: idna in /usr/local/lib/python3.12/d
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/d
Requirement already satisfied: filelock>=3.10.0 in /usr/local/lib/py
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/py
Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in /usr/local/lib/py
Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/pytl
Requirement already satisfied: jsonschema-specifications>=2023.03.0
Requirement already satisfied: referencing>=0.28.4 in /usr/local/lib/py
Requirement already satisfied: rpds-py>=0.25.0 in /usr/local/lib/py
Requirement already satisfied: six>=1.9.0 in /usr/local/lib/python3.12/c
Requirement already satisfied: python-dateutil>=2.5.3 in /usr/local/lib/py
Requirement already satisfied: websocket-client!=0.40.0,!0.41.*,!0.42.* in /usr/loca
Requirement already satisfied: requests in /usr/local/lib/python3.12/c
Requirement already satisfied: requests-oauthlib in /usr/local/lib/py
Requirement already satisfied: urllib3!=2.6.0,>=1.24.2 in /usr/local/lib/py
Requirement already satisfied: durationpy>=0.7 in /usr/local/lib/py
Requirement already satisfied: flatbuffers in /usr/local/lib/python3.12/c
Requirement already satisfied: protobuf in /usr/local/lib/python3.12/c
Requirement already satisfied: sympy in /usr/local/lib/python3.12/c
Requirement already satisfied: importlib-metadata<8.8.0,>=6.0 in /usr/local/lib/py
Requirement already satisfied: googleapis-common-protos~=1.5.7 in /usr/local/lib/py
Requirement already satisfied: opentelemetry-exporter-otlp-proto-common in /usr/local/lib/py
Requirement already satisfied: opentelemetry-proto==1.40.0 in /usr/local/lib/py
Requirement already satisfied: opentelemetry-semantic-conventions==0.45b0 in /usr/local/lib/py
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/py
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/py
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/py
Requirement already satisfied: python-dotenv>=0.21.0 in /usr/local/lib/py
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/py
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/py
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/c
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/c
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/c
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/c
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/c
```

```
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.10/site-packages/safetensors-0.4.3-cp310-cp310-linux_x86_64.whl
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.10/site-packages/click-8.2.1-py3-none-any.whl
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.10/site-packages/shellingham-1.3.0-py3-none-any.whl
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.10/site-packages/annotated_doc-0.0.2-py3-none-any.whl
Requirement already satisfied: httpx>=0.27.0 in /usr/local/lib/python3.10/site-packages/httpx-0.27.0-py3-none-any.whl
Requirement already satisfied: uvloop>=0.15.1 in /usr/local/lib/python3.10/site-packages/uvloop-0.15.1-cp310-cp310-linux_x86_64.whl
Requirement already satisfied: watchfiles>=0.20 in /usr/local/lib/python3.10/site-packages/watchfiles-0.20.0-cp310-cp310-linux_x86_64.whl
Requirement already satisfied: websockets>=10.4 in /usr/local/lib/python3.10/site-packages/websockets-10.4-cp310-cp310-linux_x86_64.whl
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.10/site-packages/joblib-1.2.0-py3-none-any.whl
```

```
import chromadb
print("ChromaDB installed successfully")
```

ChromaDB installed successfully

```
client = chromadb.Client()
collection = client.create_collection(name="rag_demo")

print("Collection created successfully")
```

Collection created successfully

```
documents = [
    "Transformers use attention mechanisms instead of RNNs.",
    "Self-attention helps capture relationships between words.",
    "Transformers allow parallel processing and faster training."
]

collection.add(
    documents=documents,
    ids=["1", "2", "3"]
)

print("Documents added successfully")
```

/root/.cache/chroma/onnx_models/all-MiniLM-L6-v2/onnx.tar.gz: 100%|█| Documents added successfully

```
query = "What is the main idea of transformers?"
```

```
results = collection.query(  
    query_texts=[query],  
    n_results=2  
)
```

```
print(results)
```

```
datas', 'documents', 'distances'], 'data': None, 'metadatas': [[None,
```

```
print("Retrieved context:", results["documents"])
```

```
Retrieved context: [['Transformers allow parallel processing and fas
```

```
print("Answer based on context:", results["documents"][0])
```

```
ing and faster training.', 'Transformers use attention mechanisms ins
```

```
context = " ".join(results["documents"][0])
```

```
print("Retrieved Context:", context)
```

```
rocessing and faster training. Transformers use attention mechanisms i
```

```
final_answer = f"Based on the retrieved context, the answer i
```

```
print(final_answer)
```

```
Based on the retrieved context, the answer is: Transformers allow pa
```

